NEET Physics Golden Set v1.0

90 curated mixed-style questions for RAG problem-solving evaluation

Instructions: This is a question-only version. Each question has four options.
Diagrams, graphs, circuits and mathematical expressions are preserved from the source question blocks.
Answers, solutions, chapter headings, difficulty labels, and original source numbering are intentionally not includ

Q1.

An electron (mass 9×10^{-31} kg and charge 1.6×10^{-19} C) moving with speed c/100 (c = speed of light) is injected into a magnetic field $\vec{B}$ of magnitude 9×10^{-4} T perpendicular to its direction of motion. We wish to apply an uniform electric $\vec{E}$ together with the magnetic field so that the electron does not deflect from its path. Then (speed of light $c = 3 \times 10^8$ ms^{-1})

(1) $\vec{E}$ is perpendicular to $\vec{B}$ and its magnitude is 27×10^4 V m^{-1}

(2) $\vec{E}$ is perpendicular to $\vec{B}$ and its magnitude is 27×10^2 V m^{-1}

(3) $\vec{E}$ is parallel to $\vec{B}$ and its magnitude is 27×10^2 V m^{-1}

(4) $\vec{E}$ is parallel to $\vec{B}$ and its magnitude is 27×10^4 V m^{-1}

Q2.

A particle of mass m is moving around the origin with a constant force F pulling it towards the origin. If Bohr model is used to describe its motion, the radius r of the nth orbit and the particle's speed v in the orbit depend on n as

(1) $r \propto n^{1/3}; v \propto n^{1/3}$

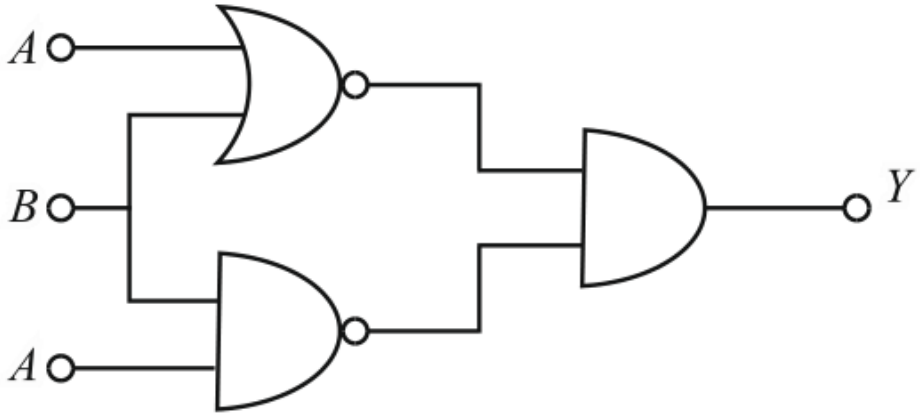
(2) $r \propto n^{1/3}; v \propto n^{2/3}$

(3) $r \propto n^{2/3}; v \propto n^{1/3}$

(4) $r \propto n^{4/3}; v \propto n^{-1/3}$

Q3.

The output (Y) of the given logic implementation is similar to the output of an/a _____gate.



(1) AND

(2) NAND

(3) OR

(4) NOR

Q4.

A microscope has an objective of focal length 2 cm, eyepiece of focal length 4 cm and the tube length of 40 cm. If the distance of distinct vision of eye is 25 cm, the magnification in the microscope is

(1) 100 (2) 125

(3) 150 (4) 250

Q5.

An oxygen cylinder of volume 30 litre has 18.20 moles of oxygen. After some oxygen is withdrawn from the cylinder, its gauge pressure drops to 11 atmospheric pressure at temperature 27°C. The mass of the oxygen withdrawn from the cylinder is nearly equal to :

[Given, $R = \frac{100}{12} J\,mol^{-1}K^{-1}$, and molecular mass of $O_2 = 32$, 1 atm pressure = $1.01 \times 10^5\, N/m^2$]

(1) 0.125 kg

(2) 0.144 kg

(3) 0.116 kg

(4) 0.156 kg

Q6.

Consider the diameter of a spherical object being measured with the help of a Vernier callipers. Suppose its 10 Vernier Scale Divisions (V.S.D.) are equal to its 9 Main Scale Divisions (M.S.D.). The least division in the M.S. is 0.1 cm and the zero of V.S. is at $x = 0.1$ cm when the jaws of Vernier callipers are closed.

If the main scale reading for the diameter is $M = 5$ cm and the number of coinciding vernier division is 8, the measured diameter after zero error correction, is:

(1) 5.18 cm (2) 5.08 cm

(3) 4.98 cm (4) 5.00 cm

Q7.

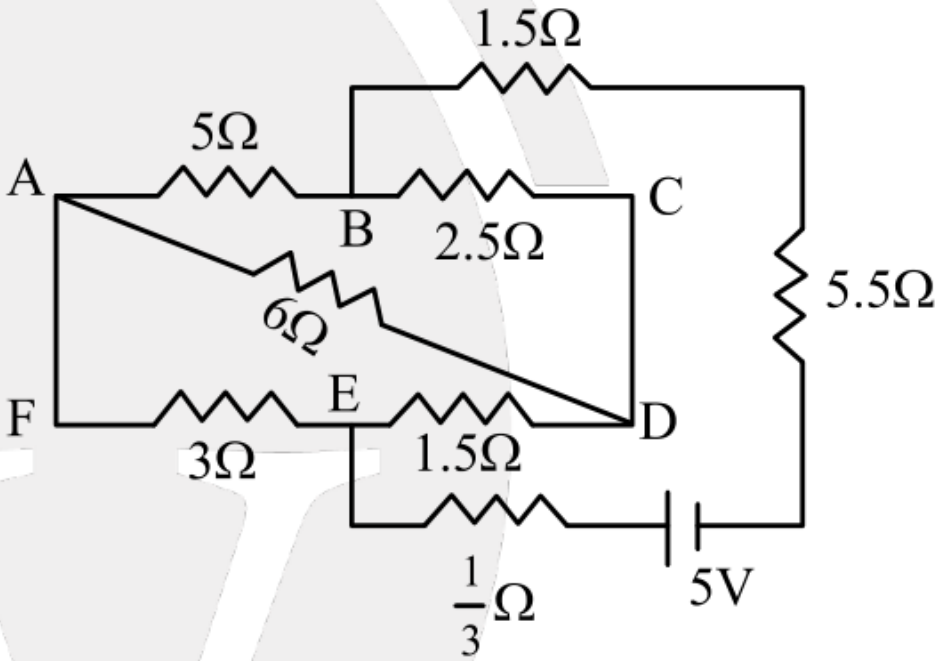
The kinetic energies of two similar cars A and B are 100 J and 225 J respectively. On applying breaks, car A stops after 1000 m and car B stops after 1500 m. If F_A and F_B are the forces applied by the breaks on cars A and B, respectively, then the ratio F_A/F_B is

(1) $\frac{3}{2}$ (2) $\frac{2}{3}$

(3) $\frac{1}{3}$ (4) $\frac{1}{2}$

Q8.

The current passing through the battery in the given circuit is:



(1) 2.0 *A*

(2) 0.5 *A*

(3) 2.5 *A*

(4) 1.5 *A*

Q9.

De-Broglie wavelength of an electron orbiting in the $n = 2$ state of hydrogen atom is close to (Given Bohr radius = 0.052 nm)

(1) 0.067 nm

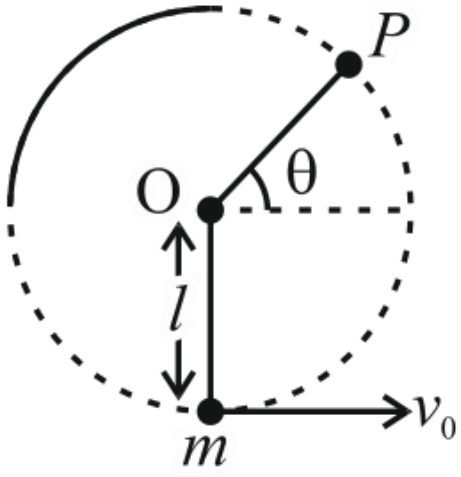
(2) 0.67 nm

(3) 1.67 nm

(4) 2.67 nm

Q10.

A bob of heavy mass m is suspended by a light string of length l. The bob is given a horizontal velocity v_0 as shown in figure. If the string gets slack at some point P making an angle θ from the horizontal, the ratio of the speed v of the bob at point P to its initial speed v_0 is:



(1) $(\sin\theta)^{1/2}$

(2) $\left(\frac{1}{2+3\sin\theta}\right)^{1/2}$

(3) $\left(\frac{\cos\theta}{2+3\sin\theta}\right)^{1/2}$

(4) $\left(\frac{\sin\theta}{2+3\sin\theta}\right)^{1/2}$

Q11.

In a certain camera, a combination of four similar thin convex lenses are arranged axially in contact. Then the power of the combination and the total magnification in comparison to the power (p) and magnification (m) for each lens will be, respectively–

(1) $4p$ and $4m$

(2) p^4 and $4m$

(3) $4p$ and m^4

(4) p^4 and m^4

Q12.

The electric field in a plane electromagnetic wave is given by

$E_z = 60 \cos (5x + 1.5 \times 10^9 t) V/m.$

Then expression for the corresponding magnetic field is (here subscripts denote the direction of the field):

(1) $B_y = 2 \times 10^{-7} \cos (5x + 1.5 \times 10^9 t) T$

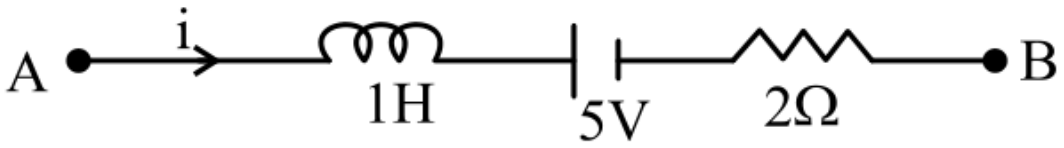
(2) $B_x = 2 \times 10^{-7} \cos (5x + 1.5 \times 10^9 t) T$

(3) $B_z = 60 \cos (5x + 1.5 \times 10^9 t) T$

(4) $B_y = 60 \sin (5x + 1.5 \times 10^9 t) T$

Q13.

AB is a part of an electrical circuit (see figure). The potential difference "$V_A - V_B$", at the instant when current i = 2 A and is increasing at a rate of 1 amp/second is:



(1) 5 volt

(2) 6 volt

(3) 9 volt

(4) 10 volt

Q14.

A ball of mass 0.5 kg is dropped from a height of 40 m. The ball hits the ground and rises to a height of 10 m. The impulse imparted to the ball during its collision with the ground is

(Take g = 9.8 m/s^2)

(1) 21 Ns

(2) 7 Ns

(3) 0

(4) 84 Ns

Q15.

A photon and an electron (mass m) have the same energy E. The ratio $\left(\lambda_{\text{photon}}/\lambda_{\text{electron}}\right)$ of their de Broglie wavelengths is (c is the speed of light)

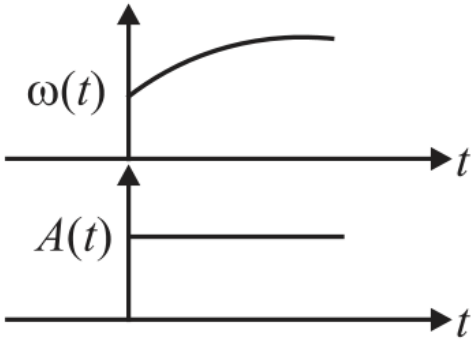
(1) $\sqrt{E/2m}$ (2) $c\sqrt{2mE}$

(3) $c\sqrt{\frac{2m}{E}}$ (4) $\frac{1}{c}\sqrt{E/2m}$

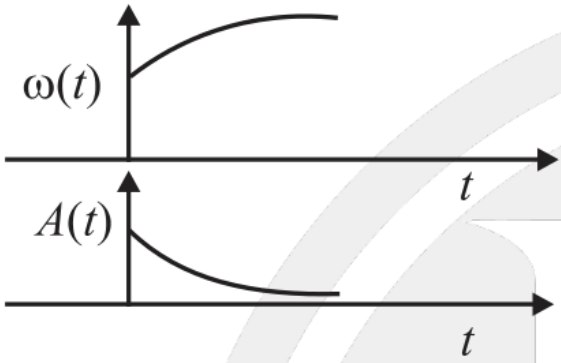
Q16.

In an oscillating spring mass system, a spring is connected to a box filled with sand. As the box oscillates, sand leaks slowly out of the box vertically so that the average frequency $\omega(t)$ and average amplitude $A(t)$ of the system change with time t. Which one of the following options schematically depicts these changes correctly?

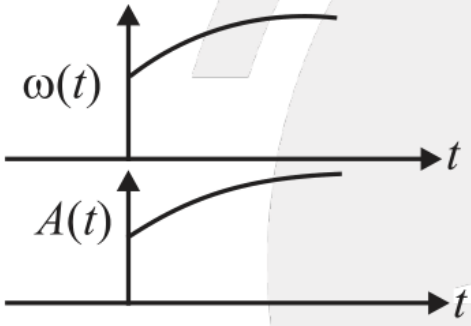
(1)



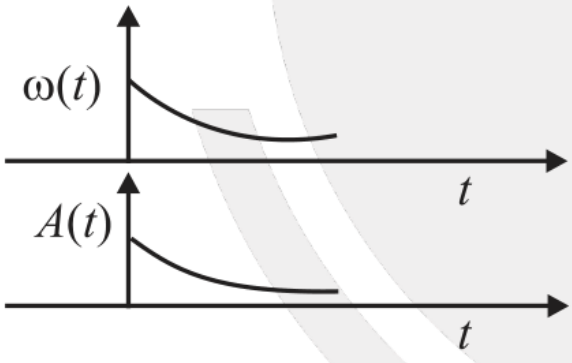
(2)



(3)



(4)



Q17.

Three identical heat conducting rods are connected in series as shown in the figure. The rods on the sides have thermal conductivity 2K while that in the middle has thermal conductivity K. The left end of the combination is maintained at temperature $3T$ and the right end at T. The rods are thermally insulated from outside. In steady state, temperature at the left junction is T_1 and that at the right junction is T_2. The ratio T_1/T_2 is:

Q18.

A physical quantity P is related to four observations a, b, c and d as follows:

$$P = a^3 b^2 / c\sqrt{d}$$

The percentage errors of measurement in a, b, c and d are 1%, 3%, 2%, and 4% respectively. The percentage error in the quantity P is

(1) 10% (2) 2%

(3) 13% (4) 15%

Q19.

In some appropriate units, time (t) and position (x) relation of a moving particle is given by $t = x^2 + x$. The acceleration of the particle is

(1) $-\frac{2}{(x+2)^3}$

(2) $-\frac{2}{(2x+1)^3}$

(3) $+\frac{2}{(x+1)^3}$

(4) $+\frac{2}{2x+1}$

Q20.

A 2 amp current is flowing through two different small circular copper coils having radii ratio 1:2. The ratio of their respective magnetic moments will be

(1) 1:4

(2) 1:2

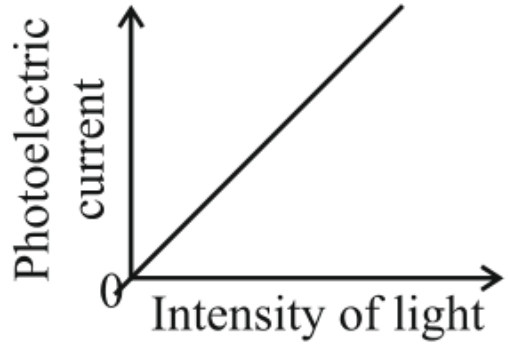
(3) 2:1

(4) 4:1

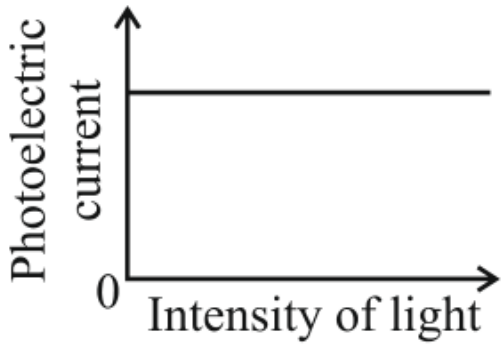
Q21.

Which of the following options represent the variation of photoelectric current with property of light shown on the x-axis?

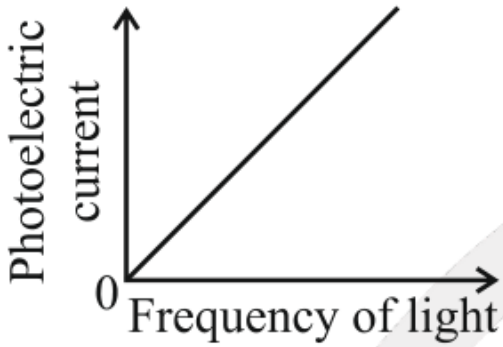
A.



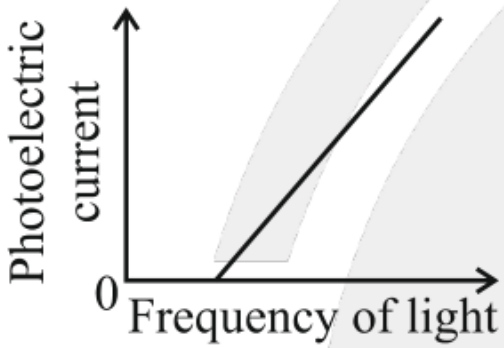
B.



C.



D.



(1) A only
(2) A and C
(3) A and D
(4) B and D

Q22.

The Sun rotates around its centre once in 27 days. What will be the period of revolution if the Sun were to expand to twice its present radius without any external influence? Assume the Sun to be a sphere of uniform density.

(1) 100 days

(2) 105 days

(3) 115 days

(4) 108 days

Q23.

An unpolarized light beam travelling in air is incident on a medium of refractive index 1.73 at Brewster's angle. Then-

(1) reflected light is completely polarized and the angle of reflection is close to 60°

(2) reflected light is partially polarized and the angle of reflection is close to 30°.

(3) both reflected and transmitted light are perfectly polarized with angles of reflection and refraction close to 60° and 30°, respectively.

(4) transmitted light is completely polarized with angle of refraction close to 30°

Q24.

To an ac power supply of 220 V at 50 Hz, a resistor of 20 Ω, a capacitor of reactance 25 Ω and an inductor of reactance 45 Ω are connected in series. The corresponding current in the circuit and the phase angle between the current and the voltage is, respectively:

(1) 7.8 A and 30°

(2) 7.8 A and 45°

(3) 15.6 A and 30°

(4) 15.6 A and 45°

Q25.

A model for quantized motion of an electron in a uniform magnetic field B states that the flux passing through the orbit of the electron is $n(h/e)$ where n is an integer, h is Planck's constant and e is the magnitude of electron's charge. According to the model, the magnetic moment of an electron in its lowest energy state will be (m is the mass of the electron)

(1) $\frac{he}{\pi m}$

(2) $\frac{he}{2\pi m}$

(3) $\frac{heB}{\pi m}$

(4) $\frac{heB}{2\pi m}$

Q26.

Two cities X and Y are connected by a regular bus service with a bus leaving in either direction every T min. A girl is driving scooty with a speed of 60 km/h in the direction X to Y notices that a bus goes past her every 30 minutes in the direction of her motion, and every 10 minutes in the opposite direction. Choose the correct option for the period T of the bus service and the speed (assumed constant) of the buses.

(1) 9 min, 40 km/h

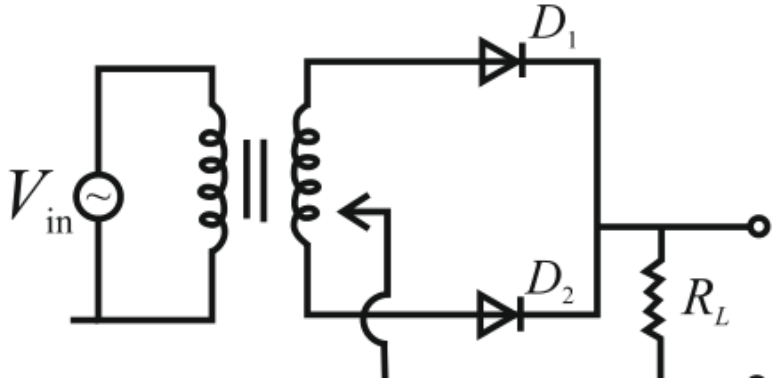
(2) 25 min, 100 km/h

(3) 10 min, 90 km/h

(4) 15 min, 120 km/h

Q27.

A full wave rectifier circuit with diodes (D_1) and (D_2) is shown in the figure. If input supply voltage $Vin = 220\sin\left(100\pi t\right)$ volt, then at t = 15 m sec



Q28.

The intensity of transmitted light when a polaroid sheet, placed between two crossed polarization at 22.5° from the polarization axis of one of the polaroid, is (I_0 is the intensity or polarised light after passing through the first polaroid):

(1) $\frac{I_0}{2}$ (2) $\frac{I_0}{4}$

(3) $\frac{I_0}{8}$ (4) $\frac{I_0}{16}$

Q29.

A container has two chambers of volumes $V_1 = 2$ litres and $V_2 = 3$ litres separated by a partition made of a thermal insulator. The chambers contains $n_1 = 5$ and $n_2 = 4$ moles of ideal gas at pressures $p_1 = 1$ atm and $p_2 = 2$ atm, respectively. When the partition is removed, the mixture attains an equilibrium pressure of:

(1) 1.3 atm

(2) 1.6 atm

(3) 1.4 atm

(4) 1.8 atm

Q30.

In a vernier calipers, $(N + 1)$ divisions of vernier scale coincide with N divisions of main scale. If 1 MSD represents 0.1 mm, the vernier constant (in cm) is:

(1) $100\,N$

(2) $10\,(N+1)$

(3) $\dfrac{1}{10N}$

(4) $\dfrac{1}{100(N+1)}$

Q31.

A uniform rod of mass 20 kg and length 5 m leans against a smooth vertical wall making an angle of 60° with it. The other end rests on a rough horizontal floor. The friction force that the floor exerts on the rod is:

(take $g = 10\ \text{m/s}^2$)

(1) 100 N

(2) $100\sqrt{3}$ N

(3) 200 N

(4) $200\sqrt{3}$ N

Q32.

The plates of a parallel plate capacitor are separated by d. Two slabs of different dielectric constant K_1 and K_2 with thickness $\frac{3}{8}d$ and $\frac{d}{2}$, respectively are inserted in the capacitor. Due to this, the capacitance becomes two times larger than when there is nothing between the plates.

If $K_1 = 1.25\ K_2$, the value of K_1 is:

(1) 2.66

(2) 2.33

(3) 1.60

(4) 1.33

Q33.

A logic circuit provides the output Y as per the following truth table:

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(1) $\overline{B}$ (2) B

(3) $A.B + \overline{A}$ (4) $A.\overline{B} + \overline{A}$

Q34.

The radius of Martian orbit around the Sun is about 4 times the radius of the orbit of Mercury. The Martian year is 687 Earth days. Then which of the following is the length of I year on Mercury?

(1) 88 earth days

(2) 225 earth days

(3) 172 earth days

(4) 124 earth days

Q35.

Two identical point masses P and Q, suspended from two separate massless springs of spring constants k_1 and k_2 respectively, oscillate vertically. If their maximum speeds are the same, the ratio (A_Q/A_P) of the amplitude A_Q of mass Q to the amplitude A_P of mass P is:

(1) $\frac{k_2}{k_1}$ (2) $\frac{k_1}{k_2}$

(3) $\sqrt{\frac{k_2}{k_1}}$ (4) $\sqrt{\frac{k_1}{k_2}}$

Q36.

A parallel plate capacitor made of circular plates is being charged such that the surface charge density on its plates is increasing at a constant rate with time. The magnetic field arising due to displacement current is:

(1) zero at all places

(2) constant between the plates and zero outside the plates.

(3) non-zero everywhere with maximum at the imaginary cylindrical surface connecting peripheries of the plates.

(4) zero between the plates and non-zero outside.

Q37.

Two identical charged conducting spheres A and B have their centres separated by a certain distance. Charge on each sphere is q and the force of repulsion between them is F. A third identical uncharged conducting sphere is brought in contact with sphere A first and then with B and finally removed from both. New force of repulsion between spheres A and B (Radii of A and B are negligible compared to the distance of separation so that for calculating force between them they can be considered as point charges) is best given as:

(1) $\frac{3F}{5}$

(2) $\frac{2F}{3}$

(3) $\frac{F}{2}$

(4) $\frac{3F}{8}$

Q38.

A body weight 48 N on the surface of the earth. The gravitational force experienced by the body due to the earth at a height equal to one-third the radius of the earth from its surface is:

(1) 16 N

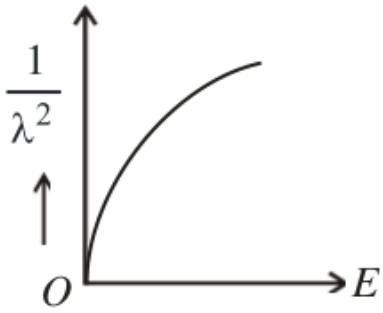
(2) 27 N

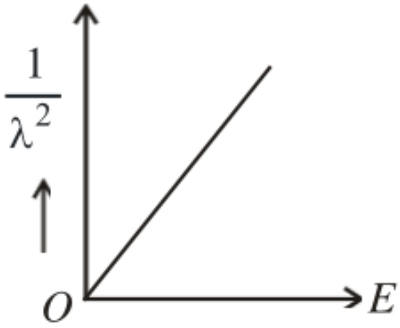
(3) 32 N

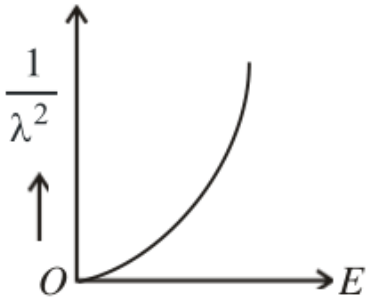
(4) 36 N

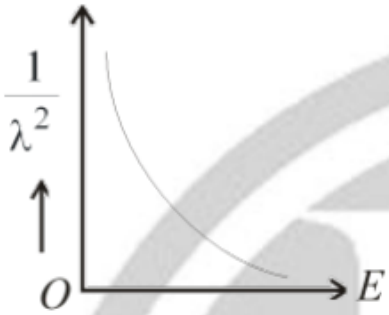
Q39.

The graph which shows the variation of $\left(\frac{1}{\lambda^2}\right)$ and its kinetic energy, E is (where λ is de Broglie wavelength of a free particle):

(1) 

(2) 

(3) 

(4) 

Q40.

A pipe open at both ends has a fundamental frequency f in air. The pipe is now dipped vertically in a water drum to half of its length. The fundamental frequency of the air column is now equal to:

(1) $\frac{f}{2}$

(2) f

(3) $\frac{3f}{2}$

(4) $2f$

Q41.

Two gases A and B are filled at the same pressure in separate cylinders with movable pistons of radius r_A and r_B, respectively. On supplying an equal amount of heat to both the systems reversibly under constant pressure, the pistons of gas A and B are displaced by 16 cm and 9 cm, respectively. If the change in their internal energy is the same, then the ratio $r_A/\ r_B$ is equal to

(1) $\frac{4}{3}$ (2) $\frac{3}{4}$

(3) $\frac{2}{\sqrt{3}}$ (4) $\frac{\sqrt{3}}{2}$

Q42.

The quantities which have the same dimensions as those of solid angle are;

(1) strain and arc

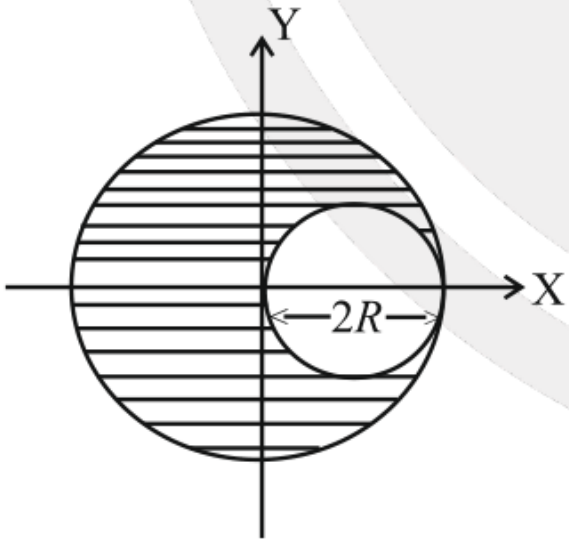
(2) angular speed and stress

(3) strain and angle

(4) stress and angle

Q43.

A sphere of radius R is cut from a larger solid sphere or radius $2R$ as shown in the figure. The ratio of the moment of inertia of the smaller sphere to that of the rest part of the sphere about the Y-axis is:



(1) $\frac{7}{8}$

(2) $\frac{7}{40}$

(3) $\frac{7}{57}$

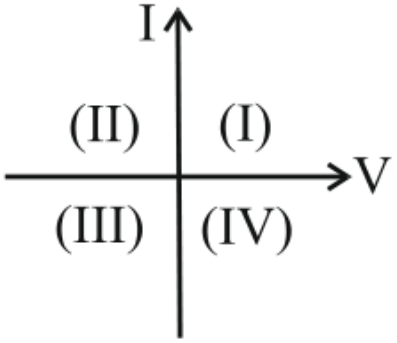
(4) $\frac{7}{64}$

Q44.

A wire of resistance R is cut into 8 equal pieces. From these pieces two equivalent resistances are made by adding four of these together in parallel. Then these two sets are added in series. The net effective resistance of the combination is:

Q45.

Consider the following statements A and B and identify the correct answer :



A. For a solar-cell, the I-V characteristics lies in the IV quadrant of the given graph.

B. In a reverse biaşed *pn* junction diode, the current measured in (μA), is due to majority charge carriers.

(1) Both A and B are correct.

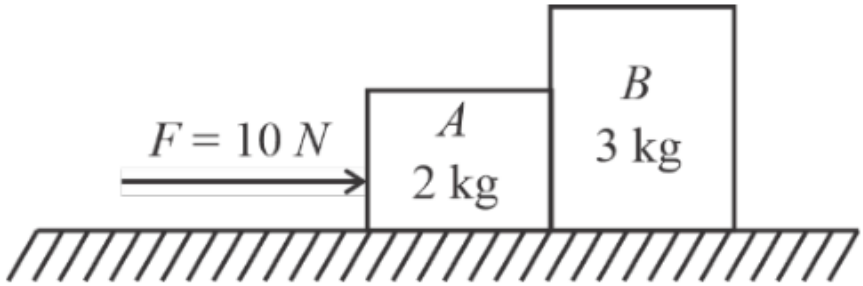
(2) Both A and B are incorrect.

(3) A is correct but B is incorrect.

(4) A is incorrect but B is correct.

Q46.

A horizontal force 10 N is applied to a block A as shown in figure. The mass of blocks A and B are 2 kg and 3 kg, respectively. The blocks slide over a frictionless surface. The force exerted by block A on block B is:



(1) 6 N (2) 10 N

(3) zero (4) 4 N

Q47.

An unpolarised light beam strikes a glass surface at Brewster's angle. Then

(1) both the reflected and refracted light will be completely polarised.

(2) the reflected light will be completely polarised but the refracted light will be partially polarised.

(3) the reflected light will be partially polarised.

(4) the refracted light will be completely polarised.

Q48.

An electric dipole with dipole moment 5×10^{-6} Cm is aligned with the direction of a uniform electric field of magnitude 4×10^{5} N/C. The dipole is then rotated through an angle of 60° with respect to the electric field. The change in the potential energy of the dipole is:

(1) 0.8 J

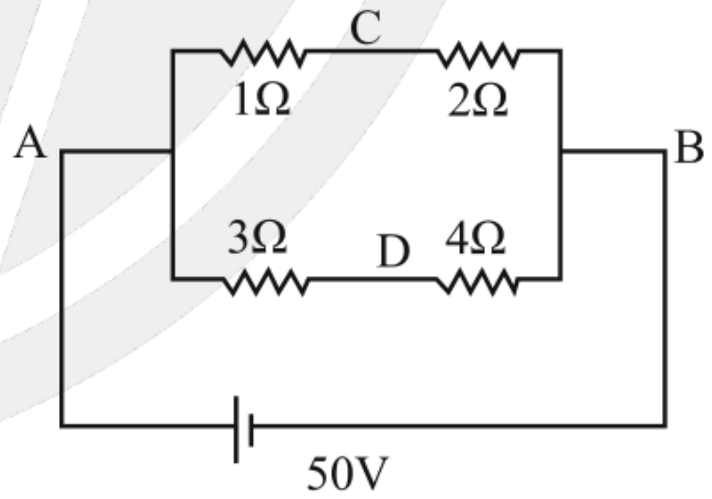
(2) 1.0 J

(3) 1.2 J

(4) 1.5 J

Q49.

A constant voltage of 50 V is maintained between the points A and B of the circuit shown in the figure. The current through the branch CD of the circuit is:



(1) 1.5 A

(2) 2.0 A

(3) 2.5 A

(4) 3.0 A

Q50.

The moment of inertia of a thin rod about an axis passing through its mid point and perpendicular to the rod, is 2400 g cm^2. The length of the 400 g rod is nearly:

(1) 20.7 cm

(2) 72.0 cm

(3) 8.5 cm

(4) 17.5 cm

Q51.

Match the **List-I** with **List-II**.

List-I (Spectral Lines of Hydrogen for transitions from)		**List-II (Wavelengths (nm))**	
(A)	$n_2 = 3$ to $n_1 = 2$	I.	410.2
(B)	$n_2 = 4$ to $n_1 = 2$	II.	434.1
(C)	$n_2 = 5$ to $n_1 = 2$	III.	656.3
(D)	$n_2 = 6$ to $n_1 = 2$	IV.	486.1

Choose the correct answer from the options given below:

(1) A-IV, B-III, C-I, D-II

(2) A-I, B-II, C-III, D-IV

(3) A-II, B-I, C-IV, D-III

(4) A-III, B-IV, C-II, D-I

Q52.

If the monochromatic source in Young's double slit experiment is replaced by white light, then

(1) there will be a central bright white fringe surrounded by a few coloured fringes.

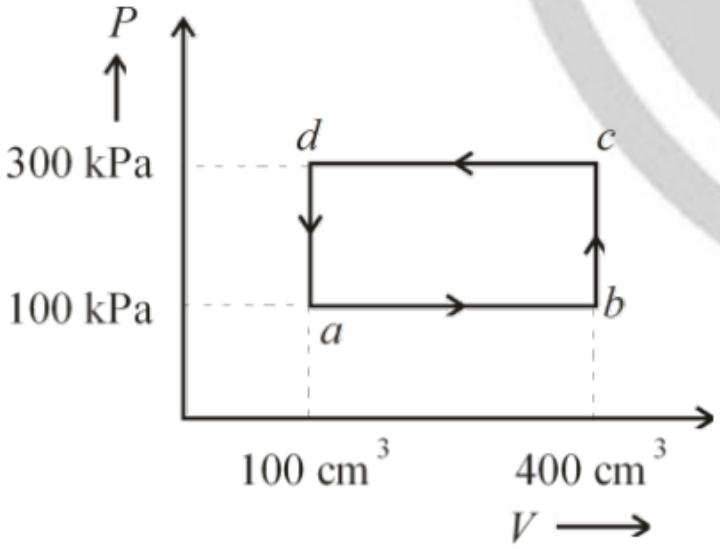
(2) all bright fringes will be of equal width.

(3) interference pattern will disappear.

(4) there will be a central dear fringe surrounded by a few coloured fringes.

Q53.

A thermodynamic system is taken through the *cycle abcda*. The work done by the gas along the path *bc* is:



(1) –90 *J*

(2) –60 *J*

(3) zero

(4) 30 *J*

Q54.

A force defined by $F= \alpha t^2+\beta t$ acts on a particle at a given time *t*. The factor which is dimensionless, if α and β are constants, is:

(1) $\alpha\beta t$

(2) $\frac{\alpha\beta}{t}$

(3) $\frac{\beta t}{\alpha}$

(4) $\frac{\alpha t}{\beta}$

Q55.

At any instant of time *t*, the displacement of any particle is given by $2t-1$ (SI unit) under the influence of force of 5*N*. The value of instantaneous power is (in SI unit):

(1) 7

(2) 6

(3) 10

(4) 5

Q56.

A tightly wound 100 turns coil of radius 10 cm carries a current of 7 A. The magnitude of the magnetic field at the centre of the coil is (Take permeability of free space as $4\pi \times 10^{-7}$ SI units):

(1) 4.4 mT
(2) 44 T
(3) 44 mT
(4) 4.4 T

Q57.

If c is the velocity of light in free space, the correct statements about photon among the following are :

A. The energy of a photon is $E = hv$.

B. The velocity of a photon is c.

C. The momentum of a photon, $p = \frac{hv}{c}$.

D. In a photon-electron collision, both total energy and total momentum are conserved.

E. Photon possesses positive charge.

Q58.

The mass of a planet is $\frac{1}{10}$ th that of the earth and its diameter is half that of the earth. The acceleration due to gravity on that planet is :

(1) $4.9\,\mathrm{m\,s^{-2}}$

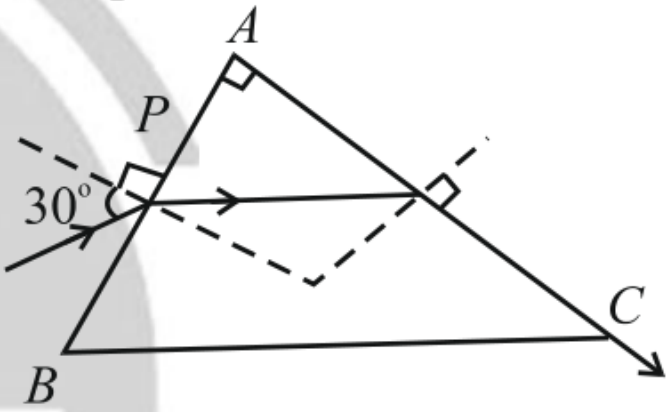
(2) $3.92\,\mathrm{m\,s^{-2}}$

(3) $19.6\,\mathrm{m\,s^{-2}}$

(4) $9.8\,\mathrm{m\,s^{-2}}$

Q59.

A light ray enters through a right angled prism at point P with the angle of incidence 30° as shown in figure. It travels through the prism parallel to its base BC and emerges along the face AC. The refractive index of the prism is:



(1) $\frac{\sqrt{3}}{4}$

(2) $\frac{\sqrt{3}}{2}$

(3) $\frac{\sqrt{5}}{4}$

(4) $\frac{\sqrt{5}}{2}$

Q60.

Math **List-I** with **List-II**.

List-I (Material)		**List-II (Susceptibility (x))**	
(A)	Diamagnetic	(I)	$x = 0$
(B)	Ferromagnetic	(II)	$0 > x \geq -1$
(C)	Paramagnetic	(III)	$x >> 1$
(D)	Non-magnetic	(IV)	$0 < x < \varepsilon$ (a small positive number)

Choose the **correct** answer from the options given below:

(1) A-III, B-II, C-I, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-III, C-IV, D-I
(4) C-II, B-I, C-III, D-IV

Q61.

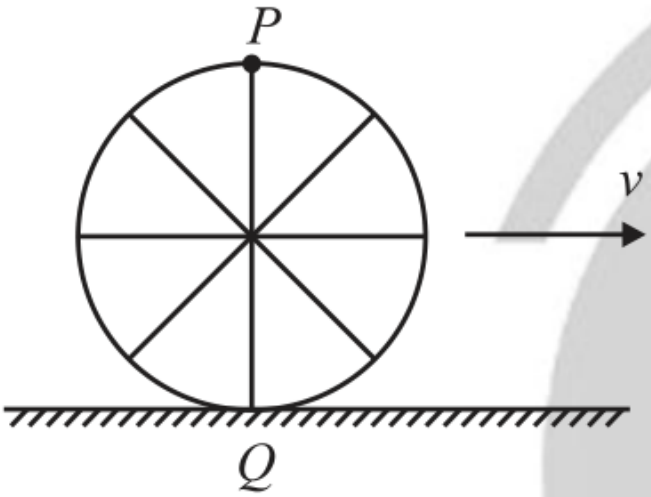
In an ideal transformer, the turns ratio is $\frac{N_p}{N_s} = \frac{1}{2}$.

The ratio $V_s : V_p$ is equal to (the symbols carry their usual meaning):

(1) 1 : 1 (2) 1 : 4
(3) 1 : 2 (4) 2 : 1

Q62.

A wheel of a bullock cart is rolling on a level road as shows in the figure below. If its linerar speed is v in the direction shown, which one of the following options is correct (P and Q are any highest and lowest points on the wheel respectively)?



(1) Both the points P and Q move with equal speed.
(2) Point P has zero speed.
(3) Point P moves slower than point Q.
(4) Point P moves faster than point Q.

Q63.

$$^{290}_{82}X \xrightarrow{\alpha} Y \xrightarrow{e^+} Z \xrightarrow{\beta^-} P \xrightarrow{e^-} Q$$

In the nuclear emission stated above, the mass number and atomic number of the product Q respectively, are:

(1) 288, 82

(2) 286, 81

(3) 280, 81

(4) 286, 80

Q64.

If $x=5\sin\left(\pi t+\frac{\pi}{3}\right)m$ represents the motion of a particle execting simple harmonic motion, the amplitude and time period of motion, respectively, are;

(1) 5 cm, 1 s

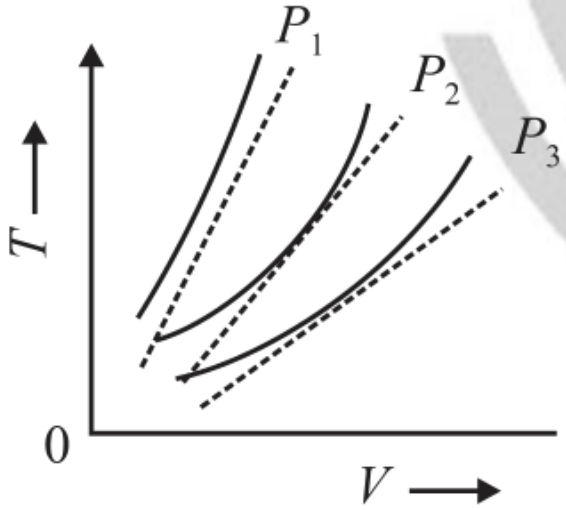
(2) 5 m, 1 s

(3) 5 cm, 2 s

(4) 5 m, 2 s

Q65.

The following graph represents the T-V curves of an ideal gas (where T is the temperature and V the volume) at three pressures P_1, P_2 and P_3 compared with those of Charles's law represented as dotted lines.



(1) $P_2 > P_1 > P_3$

(2) $P_1 > P_2 > P_3$

(3) $P_3 > P_2 > P_1$

(4) $P_1 > P_3 > P_2$

Q66.

A metal wire has mass (0.4 ± 0.002) g, radius (0.3 ± 0.001) mm and length (5 ± 0.02) cm. The maximum possible percentage error in the measurement of density will nearly be :

a. 1.4%

b. 1.2%

c. 1.3%

d. 1.6%

Motion in a Straight Line

Q67.

A particle moving with uniform speed in a circular path maintains;

(1) constant velocity but varying acceleration.

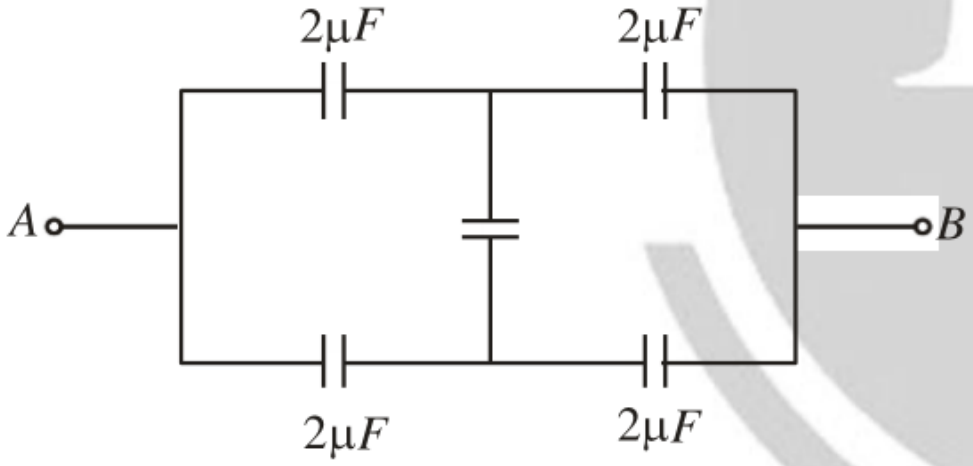
(2) varying velocity and varying acceleration.

(3) constant velocity.

(4) constant acceleration.

Q68.

In the following circuit, the equivalent capacitance between terminal A and terminal B is:



(1) 0.5 μF

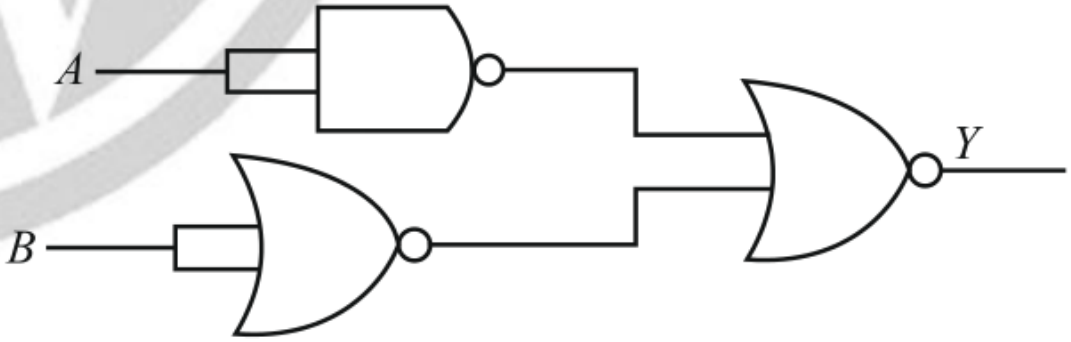
(2) 4 μF

(3) 2 μF

(4) 1 μF

Q69.

The output (Y) of the given logic gate is similar to the output of an/a :



(1) OR gate
(2) AND gate
(3) NAND gate
(4) NOR gate

Q70.

A thin flat circular disc of radius 4.5 cm is placed gently over the surface of water. If surface tension of water is 0.07 Nm^{-1}, then the excess force required to take it away from the surface is;

(1) 1.98 mN (2) 99 N
(3) 19.8 mN (4) 198 N

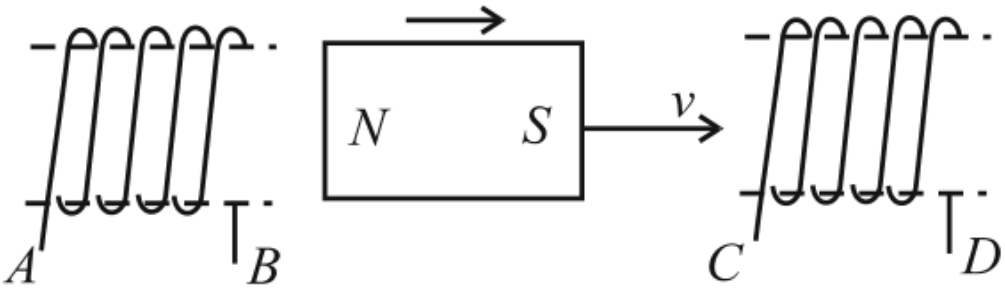
Q71.

A small telescope has an objective of focal length 140 cm and an eye piece of focal length 5.0 cm. The magnifying power of telescope for viewing a distant object is:

(1) 17 (2) 32

(3) 34 (4) 28

Q72.



Solenoid-1 Solenoid-2

In the above diagram, a strong bar magnet is moving towards solenoid-2 from solenoid-1. The direction of induced current in solenoid-1 and that in solenoid-2, respectively, are through the directions:

(1) *AB* and *CD*

(2) *BA* and *DC*

(3) *AB* and *DC*

(4) *BA* and *CD*

Q73.

Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: The potential (V) at any axial point, at 2 m distance (r) from the centre of the dipole of dipole moment vector $\vec{P}$ of magnitude, 4×10^{-6} C m, is $\pm9\times10^{3}$ V.

(Take $\frac{1}{4\pi\epsilon_0}=9\times10^{9}$ SI units)

Reason R: $V=\pm\frac{2P}{4\pi\epsilon_0 r^2}$, where r is the distance of any axial point, situated at 2 m from the centre of the dipole.

In the light of the above statements, choose the *correct* answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

Q74.

Two bodies A and B of same mass undergo completely inelastic one dimensional collision. The body A moves with velocity v_1 while body B is at rest before collision. The velocity of the system after collision is v_2. The ratio $v_1 : v_2$ is;

(1) 4 : 1 (2) 1 : 4

(3) 1 : 2 (4) 2 : 1

Q75.

Given below are two statements :

Statement I: Atoms are electrically neutral as they contain equal number of positive and negative charges.

Statement II: Atoms of each element are stable and emit their characteristic spectrum.

In the fight of the above statements, choose the *most appropriate* answer from the options given below :

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

Q76.

A Carnot engine has an efficiency of 50% when its source is at a temperature 327°C. The temperature of the sink is:

a. 200° C
b. 27° C
c. 15° C
d. 100° C

Kinetic Theory

Q77.

A bob is whirled in a horizontal plane by means of a string with an initial speed of ω rpm. The tension in the string is T. If speed becomes 2 ω white keeping the same radius, the tension in the string becomes;

(1) $\frac{T}{4}$

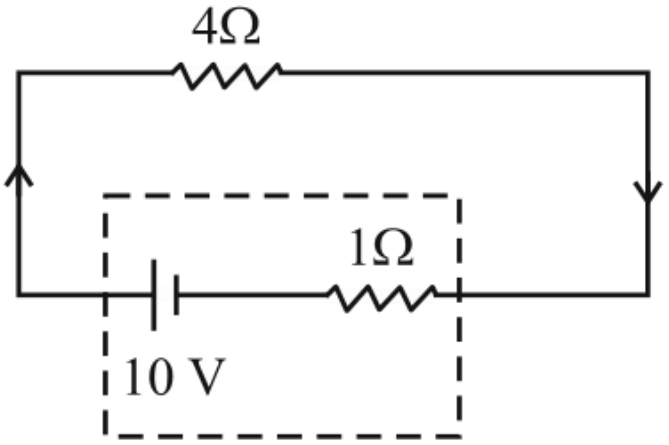
(2) $\sqrt{2}T$

(3) T

(4) $4T$

Q78.

The terminal voltage of the battery, whose emf is 10 V and internal resistance 1 Ω, when connected through an external resistance of 4 Ω as shown in the figure is :



(1) 8 *V*
(2) 10 *V*
(3) 4 *V*
(4) 6 *V*

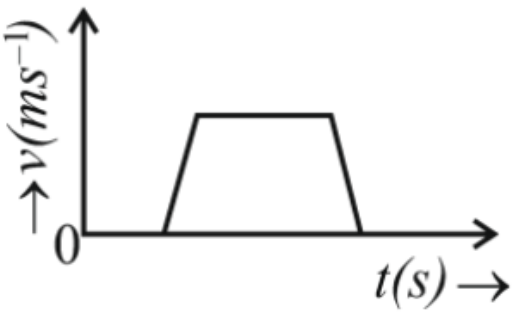
Q79.

The minimum wavelength of X-rays produced by an electron accelerated through a potential difference of V volts is proportional to:

a. V^2
b. $\sqrt{V}$
c. $\frac{1}{V}$
d. $\frac{1}{\sqrt{V}}$

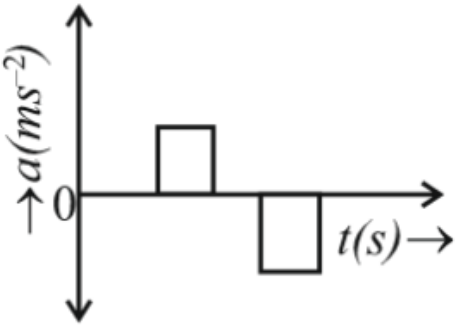
Q80.

The velocity (v) –time (t) plot of the motion of a body is shown below:

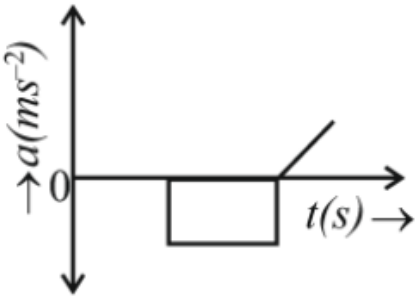


The acceleration (a) –time (t) graph that best suits this motion is:

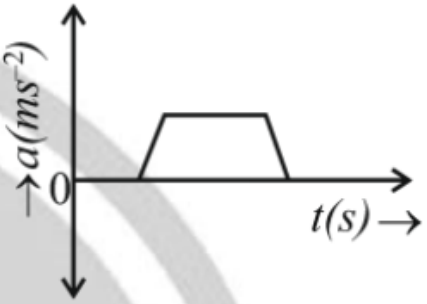
(1)



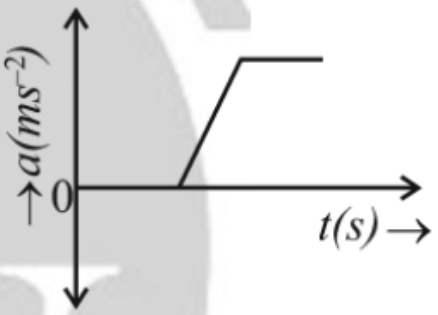
(2)



(3)



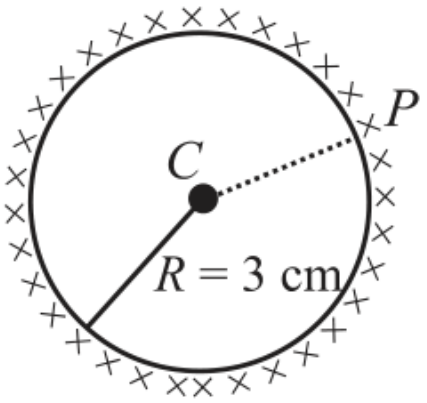
(4)



Q81.

A thin spherical shell is charged by some source. The potential difference between the two points C and P (in V) shown in the figure is;

$$\left(\text{Take}\,\frac{1}{4\pi\in_0}=9\times10^9\ \text{SI units}\right)$$



$q = 1\mu C$

(1) 0.5×10^5

(2) zero

(3) 3×10^5

(4) 1×10^5

Q82.

A parallel plate capacitor is charged by connecting it to a battery through a resistor. If I is the current in the circuit, then in the gap between the plates;

(1) displacement current of magnitude equal to I flows m a direction opposite to that of I.

(2) displacement current of magnitude greater than I flows but can be in any direction.

(3) there is no current.

(4) displacement current of magnitude equal to I flows in the same direction as I.

Q83.

The minimum energy required to launch a satellite of mass m from the surface of earth of mass M and radius R in a circular orbit at an altitude of $2R$ from the surface of the earth is:

(1) $\frac{GmM}{2R}$ (2) $\frac{GmM}{3R}$

(3) $\frac{5GmM}{6R}$ (4) $\frac{2GmM}{3R}$

Q84.

In hydrogen spectrum, the shortest wavelength in the Balmer series is λ. The shortest wavelength in the Bracket series is :

a. 16 λ
b. 2 λ
c. 4 λ
d. 9 λ

Nuclei

Q85.

The temperature of a gas is –50 °C. To what temperature the gas should be heated so that the rms speed is increased by 3 times?

a. 223 K
b. 669 °C
c. 3295 °C
d. 3097 K

Waves

Q86.

If the plates of a parallel plate capacitor connected to a battery are moved close to each other, then

A. the charge stored in it, increase.

B. the energy stored in it, decreases.

C. its capacitance increases.

D. the ratio of charge to its potential remains the same.

E. the product of charge and voltage increases.

Choose the most appropriate answer from the options given below:

(1) B, D and E only (2) A, B and C only

(3) A, B and E only (4) A, C and E only

Q87.

The half life of a radioactive substance is 20 minutes. In how much time, the activity of substance drops to $(1/16)^{th}$ of its initial value?

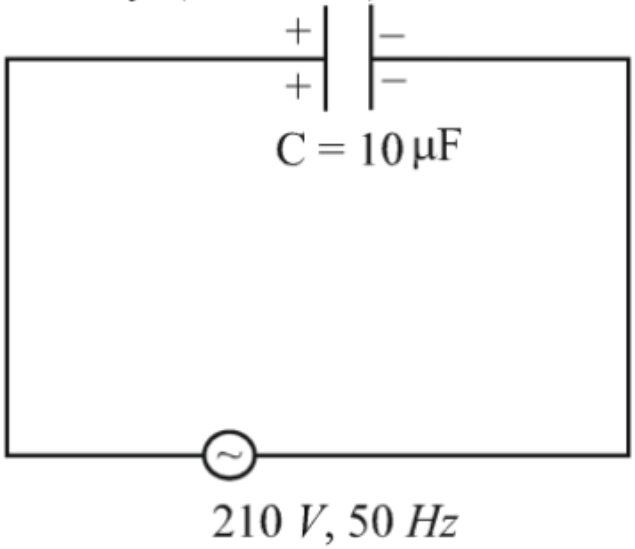
a. 80 minutes b. 20 minutes

c. 40 minutes d. 60 minutes

Semiconductor Electronics

Q88.

A 10 μF capacitor is connected to a 210 V, 50 Hz source as shown in figure. The peak current in the circuit is nearly ($\pi = 3.14$):



(1) 1.20 A
(2) 0.35 A
(3) 0.58 A
(4) 0.93 A

Q89.

A full wave rectifier circuit consists of two p-n junction diodes, a centre-tapped transformer, capacitor and a load resistance. Which of these components remove the ac ripple from the rectified output

a. Load resistance
b. A centre-tapped transformer
c. p-n junction diodes
d. Capacitor

Q90.

Given below are two statements:

Statement I: Photovoltaic devices can convert optical radiation into electricity.

Statement II: Zener diode is designed to operate under reverse bias in breakdown region.

In the light of the above statements, choose the most appropriate answer from the options given below:

a. Statement I is incorrect but Statement II is correct.

b. Both Statement I and Statement II are correct.

c. Both Statement I and Statement II are incorrect.

d. Statement I is correct but Statement II is incorrect.